# Ashvin Ranjan

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## Education

**University of Washington** | Expected September 2027 **GPA: 3.98/4.0**

*Bachelor of Science in Computer Science, Minors in Linguistics & Japanese*

- Shallow Processing Techniques for NLP (Master's course), Advanced Statistical Methods in NLP (Master's course), Introduction To Compiler Design, Introduction to Operating Systems, Modern Concurrency, Introduction to Algorithms, System Programming, Hardware/Software Interface, Data Structures & Algorithms
- Software Engineering Career Club Software Engineering Officer: 2025 - 2026, 2026 - 2027

## Skills

Languages: Python, Rust, Typescript, Javascript, C, C++, Go, Java, C#,
Next.js and React Frontend development | Typescript, Rust, and Go backend development
Research in Computer Science and Linguistics

## Work

**CLMBR** | Prof. Steinert-Threlkeld's Computational Linguistics research lab.

*Undergraduate Researcher* *September 2026 - September 2027*

**Google** |YouTube app performance and logging team.

*Software Engineering Intern* *June 2026 - Present*

- Collaborating with other teams on agentic dead code elimination along with **compiler optimizations** for dead code.
- Developing agentic and algorithmic systems to process logging and trace information for developer understanding at large scales.

**CLMBR** | Prof. Steinert-Threlkeld's Computational Linguistics research lab.

*Undergraduate Researcher* *November 2024 - June 2026*

- First author on research using a custom **numpy** and **pandas**-based **python** library to analyze Information Bottleneck systems.
- Leading research on regular expression matching at terabyte scale utilizing custom **Rust** programs, algorithms, and data structures.
- Second author on research fine-tuning and analyzing various **Hugging Face** transformers for many languages and forms of data.

**Inspirogram** | A Computer Science education non-profit.

*Software Engineering Intern* *May 2022 - January 2023*

- Volunteered to help develop the website and write courses for the non-profit, which would be used to teach students in Tanzania.
- Utilized **Next.js** to develop the website. Wrote courses on **Unity** and **React.js** software development and made presentations.

**Outreach** | A marketing and outreach company.

*Software Engineering Intern* *July 2022 - August 2022*

- Utilized **Go** to develop a backend and command line interface to create company GitHub repositories with custom templates.
- Translated Ruby code into **Go** and created a custom YAML specification to allow for the creation of custom repository templates.

*Software Engineering Intern* *June 2021 - August 2021*

- Collaborated with the calendaring experiences team in order to solve bugs in the calendar of the **Next.js**-based company website.
- Created outline for additional sections of the company website involving image processing and uploading by users using **Next.js**.

**Backbone** | A mobile gaming peripherals company.

*Software Engineering Intern* *March 2020 - August 2020*

- Created an iOS application with **Swift** to send live stream data to a computer via USB to an OBS plugin for recording or streams.

## Publications

### Preprints

Jingnong Qu, Ashvin Ranjan, & Shane Steinert-Threlkeld. (2026). Brain Score Tracks Shared Properties of Languages: Evidence from Many Natural Languages and Structured Sequences. *arXiv*. https://arxiv.org/abs/2604.15503

### 2026

Ashvin Ranjan & Shane Steinert-Threlkeld. (2026). When Efficient Communication Explains Convexity. *CogSci*.